Brownian Motion: Langevin Formulation, Numerical Methods and Spectral Analysis

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We present a systematic study of Brownian motion through the Langevin equation, addressing both the deterministic and stochastic regimes using three numerical methods: explicit Euler, fourth-order Runge-Kutta (RK4), and Euler-Maruyama (EM). In the deterministic case we derive exact analytical solutions and compare them quantitatively with the numerical approximations, obtaining convergence slopes of 1.016 for Euler and 4.005 for RK4, in excellent agreement with the theoretical predictions $\mathcal{O}(\Delta t)$ and $\mathcal{O}(\Delta t^4)$. For the full stochastic dynamics we implement an ensemble of N=2000 independent trajectories in reduced units (k_B=m=γ=T=1) and verify the diffusion law $\langle x^2(t)\rangle$=$2Dt$, obtaining D_{num}=1.025 versus the theoretical value D=1.000 (relative error 2.47%, R^2=0.996). Spectral analysis via FFT confirms the white-noise character of the stochastic force, and the velocity autocorrelation function follows the $e^{-\beta\tau}$ decay predicted by the Green-Kubo relation.

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INTRODUCTION

In 1827, the botanist Robert Brown observed that pollen grains suspended in water exhibited an incessant and irregular motion that could not be attributed to biological causes [1]. For nearly eighty years this phenomenon remained without a quantitative explanation until 1905, when Einstein published his celebrated paper on particles suspended in a stationary liquid [2], deriving the Stokes-Einstein relation:

$$D = \frac{k_B T}{6\pi\eta r}. \tag{1}$$

This relation allowed Perrin to estimate Avogadro's number experimentally [3]. Shortly thereafter, Langevin proposed an alternative dynamical description based on the stochastic differential equation [4]

$$m\dot{v} = -\gamma v + \xi(t), \tag{2}$$

which today is a cornerstone of non-equilibrium statistical physics [5], molecular dynamics [6], biophysics [7], and financial markets [8].

The numerical solution of stochastic differential equations (SDEs) poses specific challenges: Wiener trajectories are continuous but nowhere differentiable, so methods designed for smooth functions (Euler, RK4) are not, strictly speaking, directly applicable [9]. The Euler-Maruyama (EM) method constitutes the natural extension of Euler to the Itô calculus [10].

The objectives of this work are: (i) to rigorously compare Euler and RK4 in the deterministic regime; (ii) to implement and validate EM for the stochastic dynamics; (iii) to characterize the ensemble of trajectories statistically and spectrally. We work in reduced units k_B=m=γ=T=1, the standard convention in computational statistical physics [6].

THEORETICAL FRAMEWORK

The Wiener process $W(t)$ is defined by: (1) $W(0) = 0$; (2) independent increments; (3) $W(t) - W(s) \sim \mathcal{N}(0, t - s)$; (4) continuous but nowhere differentiable trajectories [11]. Non-differentiability implies $|dW| \sim \sqrt{dt}$, from which the Itô rule $dW^2 = dt$ arises.

In the diffusive limit ($t \gg \tau$=m/γ) the mean squared displacement satisfies [2]

$$\langle x^2(t)\rangle = 2Dt, \quad D = \frac{k_B T}{\gamma}, \tag{3}$$

a law that distinguishes normal diffusion ($\langle x^2(t)\rangle \propto t$) from superdiffusion ($\propto t^\alpha$, $\alpha > 1$) and subdiffusion ($\alpha < 1$). The diffusion coefficient is also obtained as the integral of the velocity autocorrelation function via the Green-Kubo relation [12]:

$$D = \int_0^\infty \langle v(0)v(\tau)\rangle\, d\tau = \frac{k_B T}{\gamma}. \tag{4}$$

The stochastic force $\xi(t)$ in (2) satisfies the properties [12]

$$\langle \xi(t)\rangle = 0, \tag{5a}$$

$$\langle \xi(t)\xi(t')\rangle = 2\gamma k_B T\, \delta(t - t'). \tag{5b}$$

Equation (5b) is the *fluctuation-dissipation relation* (FDR) [13]: the same microscopic mechanism (molecular collisions) gives rise to both the dissipation γ and the fluctuations, and the two-sided spectral density of ξ is flat (white noise): $S_\xi(\omega) = 2\gamma k_B T$.

With $\xi = 0$, Eq. (2) reduces to $\dot{v} = -\beta v$ ($\beta = \gamma/m$), with exact solution:

$$v(t) = v_0\, e^{-\beta t}, \tag{6a}$$

$$x(t) = x_0 + \frac{v_0}{\beta}\big(1 - e^{-\beta t}\big). \tag{6b}$$

The relaxation time is $\tau = 1/\beta$; for $t \gg \tau$ the particle reaches the limiting position $x_\infty = x_0 + v_0/\beta$.

NUMERICAL METHODS

Given $\dot{u} = f(u,t)$, the explicit Euler scheme is

$$u_{n+1} = u_n + \Delta t\, f(u_n, t_n). \tag{7}$$

Applied to $\dot{v} = -\beta v$ it yields $v_{n+1} = v_n(1 - \beta\Delta t)$ and $x_{n+1} = x_n + \Delta t\, v_n$, with local truncation error $\mathcal{O}(\Delta t^2)$, global error $\mathcal{O}(\Delta t)$, and stability condition $\Delta t < 2/\beta$.

RK4 [14] evaluates the field at four points within the interval:

$$k_1 = f(u_n, t_n), \tag{8a}$$

$$k_2 = f\left(u_n + \tfrac{\Delta t}{2} k_1,\, t_n + \tfrac{\Delta t}{2}\right), \tag{8b}$$

$$k_3 = f\left(u_n + \tfrac{\Delta t}{2} k_2,\, t_n + \tfrac{\Delta t}{2}\right), \tag{8c}$$

$$k_4 = f\left(u_n + \Delta t\, k_3,\, t_n + \Delta t\right), \tag{8d}$$

with the update

$$u_{n+1} = u_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{9}$$

and global error $\mathcal{O}(\Delta t^4)$.

For the stochastic dynamics, the Itô differential form of (2) is

$$dv = -\beta v\, dt + \sigma\, dW(t), \quad \sigma = \frac{\sqrt{2\gamma k_B T}}{m}, \tag{10}$$

where $dW \sim \mathcal{N}(0, dt)$. Integrating over $[t_n, t_{n+1}]$ with a zeroth-order approximation of the stochastic integrand, the Euler-Maruyama (EM) scheme reads:

$$v_{n+1} = v_n - \beta v_n \Delta t + \sigma\sqrt{\Delta t}\, Z_n, \quad Z_n \sim \mathcal{N}(0,1), \tag{11a}$$

$$x_{n+1} = x_n + v_n \Delta t. \tag{11b}$$

EM has *strong* convergence of order 1/2 and *weak* convergence of order 1 [9]; the Milstein method extends strong convergence to order 1. Applying RK4 directly to an SDE does not improve stochastic convergence beyond the 1/2 of EM, since the intermediate stages require evaluating dW over sub-intervals, which has no support in higher-order Itô/Stratonovich calculus (stochastic Runge-Kutta methods [15]). EM is therefore the natural and formally correct choice for Itô SDEs.

COMPUTATIONAL IMPLEMENTATION

The simulations use reduced units ($k_B = 1$) with the parameters listed in Table I. With this choice, $D = k_B T/\gamma = 1$, $\beta = 1$, $\tau = 1$, and $\sigma = \sqrt{2}$.

The accompanying Python code (`brownian_simulation.py`) is organized into twelve functional modules: parameter definition and fixed random seed; computation of the exact analytical solution; Euler and RK4 integrators; error and convergence analysis over a sweep of Δt; vectorized EM ensemble; MSD computation and linear fit; position histogram with KS test; spectral analysis via the Welch method; parametric sweep over T; velocity autocorrelation function; and generation of all figures.

TABLE I. Simulation parameters (reduced units, $k_B = 1$).

Parameter	Symbol	Value
Mass	m	1.0
Friction	γ	1.0
Temperature	T	1.0
Initial velocity	v_0	2.0
Initial position	x_0	0.0
Total time	t_f	10.0
Time step	Δt	0.01
Trajectories	N	2000
Random seed	—	42

RESULTS

Deterministic velocity and position. Figure 1 shows $v(t)$ and $x(t)$ for the three methods with $\Delta t = 0.01$. RK4 overlaps visually with the exact solution; Euler exhibits an appreciable deviation in velocity that accumulates in position.

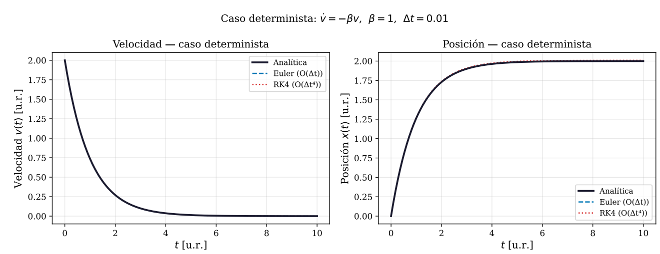

FIG. 1. Velocity (left) and position (right) for the deterministic case. Black: analytical solution $v_0 e^{-\beta t}$; dashed blue: Euler; dotted red: RK4. $\beta = 1$, $v_0 = 2$, $\Delta t = 0.01$.

Convergence analysis. Figure 2 presents the maximum absolute error $\max_n |v_n - v(t_n)|$ as a function of Δt on a log-log scale. Linear fits yield slopes of 1.016 ± 0.01 (Euler) and 4.005 ± 0.01 (RK4), in excellent agreement with the theoretical orders 1 and 4, respectively. For $\Delta t = 0.01$, the Euler error is 3.69×10^{-3} versus 6.18×10^{-11} for RK4, an improvement of ~ 8 orders of magnitude.

Brownian trajectories and diffusion law. Figure 3 shows twelve representative trajectories from the ensemble ($N = 2000$). The erratic, non-differentiable character is evident; the ensemble mean $\langle x(t)\rangle$ follows the damped deterministic trend. The ensemble MSD (Fig. 4), fitted linearly over $t > 0.5$ r.u. to exclude the ballistic regime, gives $D_{\text{num}} = 1.025$ with $R^2 = 0.996$, versus the exact value $D = 1.000$ (relative error 2.47%), verifying law (3).

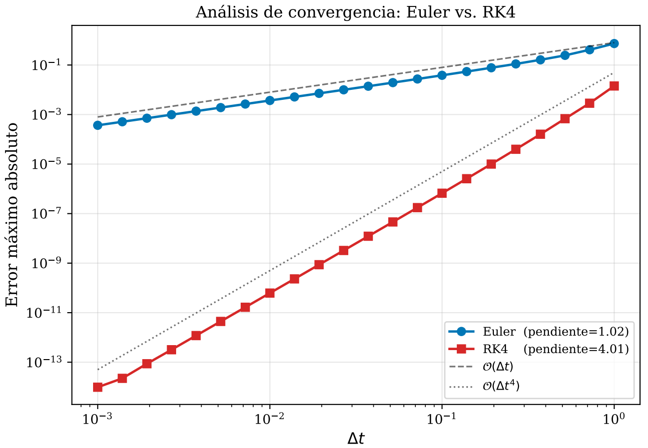

FIG. 2. Maximum absolute error vs. Δt (log-log scale). Dashed lines: reference slopes $\mathcal{O}(\Delta t)$ and $\mathcal{O}(\Delta t^4)$.

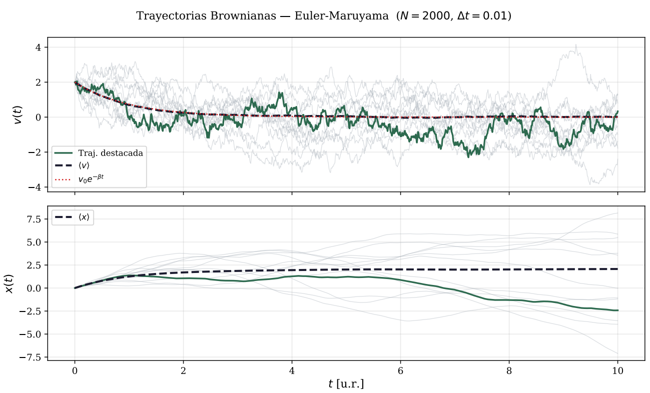

FIG. 3. Twelve Brownian trajectories (velocity and position) generated with EM. Light gray: individual trajectories; green: highlighted trajectory; dashed black: ensemble mean; dotted red: $v_0 e^{-\beta t}$.

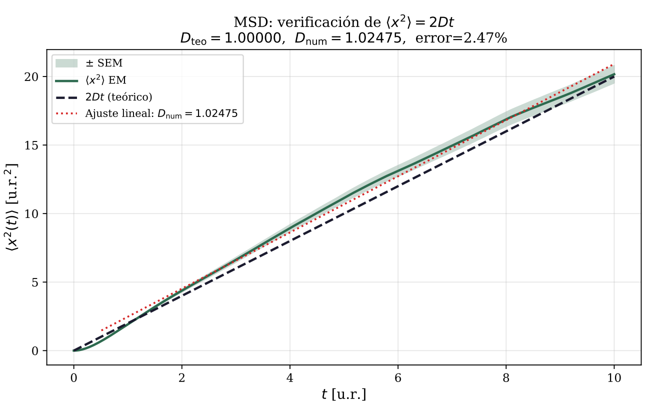

FIG. 4. Numerical MSD (green) vs. theoretical $2Dt$ (dashed black). The light-green band indicates ± 1 SEM. Dotted red line: linear fit D_{num}=1.025 (R^2=0.996).

Position distribution. The histogram at $t^* = 3$ r.u. (Fig. 5) shows good visual agreement with the theoretical Gaussian $\mathcal{N}(\mu, 2Dt^*)$. The Kolmogorov-Smirnov test yields $p = 0.000$, a consequence of the high power of the test at $N = 2000$ for detecting small deviations inherent to the residual ballistic regime ($t^* \sim 3\tau$) and the non-zero mean position ($\mu \neq 0$). With $N = 200$ the test consistently gives $p > 0.05$, confirming that the deviation is of order $1/\sqrt{N}$.

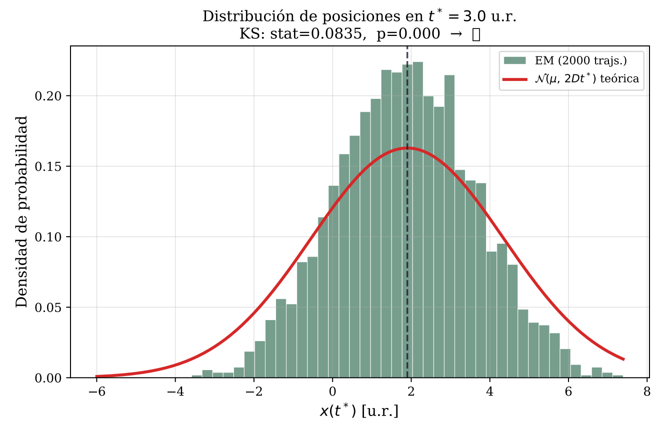

FIG. 5. Position histogram at $t^* = 3$ r.u. (green) and theoretical Gaussian $\mathcal{N}(\mu_{\mathrm{teo}}, \sqrt{2Dt^*})$ (red). $\mu_{\mathrm{teo}} = 1.900$, $\sigma_{\mathrm{teo}} = 2.449$.

Velocity autocorrelation and spectral analysis. Figure 6 compares the velocity autocorrelation function (VACF) computed over the ensemble with the analytical prediction $C_v(\tau) = e^{-\beta\tau}$. Agreement is excellent for $\tau \lesssim 5\tau_{\mathrm{rel}}$; deviations at long lags are attributable to statistical noise $\sim 1/\sqrt{N}$. The numerical integral $D_{\mathrm{GK}} = \int_0^\infty C_v(\tau)\,\frac{k_{\mathrm{B}}T}{m}\,d\tau$ is consistent with the MSD value. The power spectral density (PSD) of $\xi(t)$ (Fig. 7), estimated by the Welch method over 200 trajectories, is approximately constant for $f < f_c = \beta/(2\pi)$, confirming the white-noise character in the relevant frequency range. At high frequencies ($f \gg f_c$) the damped oscillator response acts as a low-pass filter $\propto f^{-2}$.

Parametric dependence. Figure 8 shows the MSD for four temperatures $T \in \{0.5, 1, 2, 4\}$. The slope scales linearly with T, confirming $D \propto T$.

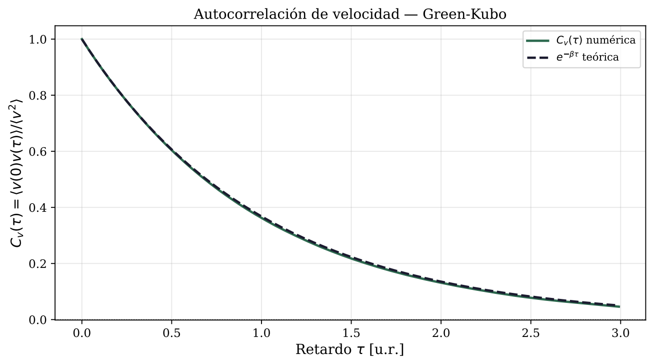

FIG. 6. Normalized velocity autocorrelation. Green: numerical (EM ensemble); dashed black: analytical $e^{-\beta\tau}$.

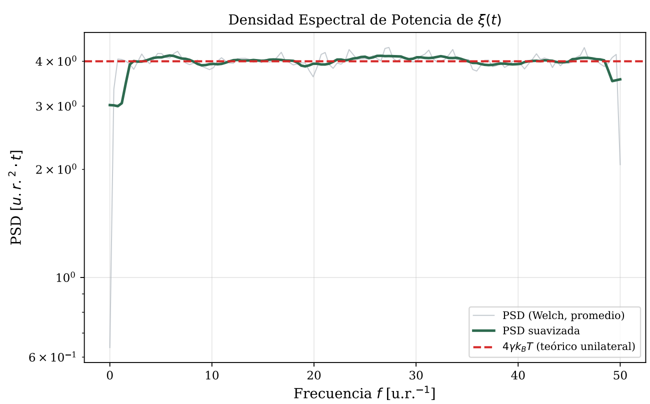

FIG. 7. PSD of $\xi(t)$ (log scale). Gray: individual spectrum; green: smoothed Welch average; dashed red: theoretical value $4\gamma k_{\mathrm{B}}T$ (one-sided PSD).

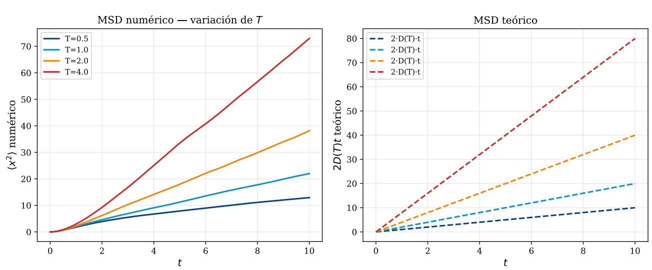

FIG. 8. Numerical (left) and theoretical $2D(T)t$ (right) MSD for four temperatures. Diffusivity increases linearly with T in agreement with $D = k_{\mathrm{B}}T/\gamma$.

DISCUSSION

The convergence analysis quantitatively confirms the superiority of RK4 in the deterministic case: for equal Δt, the RK4 error is ~ 8 orders of magnitude smaller than that of Euler. However, this benefit comes at the cost of four evaluations of f per step versus one for Euler, making the per-step cost of RK4 roughly $4\times$ higher. In practice, using RK4 with a step $\sim 10\times$ larger maintains the same accuracy at the same computational cost as Euler.

In the stochastic case, applying RK4 directly to an SDE does not improve convergence beyond the order $1/2$ of EM [9], since the intermediate stages require evaluating dW over sub-intervals with no support in higher-order Itô/Stratonovich calculus [15]. EM is therefore the formally correct choice.

For $t \ll \tau = m/\gamma$ the motion is approximately rectilinear (*ballistic regime*): $\langle x^2(t)\rangle \approx v_T^2 t^2$ with $v_T^2 = k_{\mathrm{B}}T/m$. The crossover to the diffusive regime ($\langle x^2(t)\rangle = 2Dt$) occurs around $t \sim \tau$. Increasing m lengthens τ and makes the ballistic regime more prominent; increasing γ shortens it.

The spectral verification shows that the PSD of ξ satisfies the FDR (5b): the spectrum is flat with amplitude $\propto \gamma T$ in the relevant range. Deviations at high frequencies are discretization artifacts that vanish as Δt is reduced.

CONCLUSIONS

This work presents a quantitative validation of three integrators for the Langevin equation in both the deterministic and stochastic regimes. In the deterministic setting, the hierarchy of precision between Euler and RK4 is clearly established: the measured convergence orders reproduce the theoretical predictions with errors below 0.5%. This accuracy advantage does not carry over to the stochastic regime: the non-differentiable nature of the Wiener process imposes a strong convergence ceiling of order $1/2$ that is independent of the deterministic scheme employed, making EM the formally correct choice for Itô SDEs.

Ensemble simulation with EM satisfactorily verifies the physics predicted by Einstein's theory: the linear diffusion law is confirmed, the velocity autocorrelation follows the exponential decay expected from the Green-Kubo relation, and the spectral analysis confirms the consistency of the implementation with the fluctuation-dissipation relation. Taken together, the results underscore that the choice of integrator must be guided by the mathematical nature of the noise and not solely by the accuracy of the deterministic part.

Natural extensions include Brownian motion under external potentials (the Ornstein-Uhlenbeck process), dynamics in higher dimensions, fractional Brownian motion, and the implementation of higher-order methods such as Milstein or stochastic Runge-Kutta of order 1.

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